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Chapter 13
Advancements and challenges of using natural language processing in the healthcare sector

Abstract: Due to the exponential increase in population, the healthcare sector is also booming. The data generated from the healthcare sector is too huge nowadays, and processing all of it manually is a tedious task. The major types of data, which we want to process in the healthcare sector include discharge summaries, reports from radiologists or pathologists, case notes of the physicians, etc. Nowadays, mostly, all these details are available in digital form, stored in some of the healthcare systems. In most cases, all these reports will be in an unstructured form and processing it using conventional computer algorithms is not efficient. Linguistics and computer science knowledge is required for natural language processing (NLP). The usage of NLP in the healthcare sector is widespread, allowing us to infer the meaning of unstructured healthcare information and translate it into a form that the electronic healthcare system can understand. This will help to reduce the manpower requirement in the healthcare industry, enable more structured details to be stored and reused in future for reference, and reduce the chances of human errors while processing. This manuscript gives an overview of NLP and its various phases, the various applications of NLP, the key areas where NLP can use in the healthcare sector, various well-known NLP-based approaches available for processing healthcare information, and the various datasets available for performing research in the domain of NLP on healthcare information. This manuscript also gives a brief idea about the research challenges in this domain, which need future attention.

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13.1 Introduction

Natural language processing (NLP) research has been going on since the late 1940s and has spanned several decades. The first identifiable NLP application was developed in 1948 at Birkbeck College London [1]. The very first computer-based application connected to natural language was machine translation (MT). At that time, the primary languages used were English and Russian. Initial MT research posited that language differences primarily lay in vocabulary and permissible word sequences. These systems relied solely on dictionary lookups for translation, rearranging words post-translation to align with the target language's syntactic norms, and overlooking inherent lexical ambiguities, leading to unfavorable results [1]. The seeming failure helped researchers recognize that the endeavor was far more complex than they had imagined, and they needed a better language theory. Noam Chomsky was one of the foremost researchers in developing NLP. He found it perplexing because language models treated nonsensical but grammatically accurate and unimportant phrases as nonsense but not the grammatically right ones. The field gained a clearer understanding of whether or how mainstream linguistics might benefit from MT when Noam Chomsky released Syntactic Structures in 1957, introducing the notion of generative grammar.

Between 1957 and 1970, researchers in the field of NLP broke into two groups: symbolic and stochastic. In symbolic, researchers worked on formal languages and syntax generation; this group included linguists and computer scientists who saw this section also as the start of artificial intelligence research. Whereas various statistical NLP approaches piqued the interest of stochastic researchers, who worked on challenges like optical character recognition and pattern detection across texts. As more technology and information became accessible after 1970, researchers splintered even further, adopting new areas of NLP. Logic-based paradigms, or languages that concentrated on encoding rules and language in mathematical logics, were a new topic. This branch of NLP study later facilitated the emergence of Prolog, a computer language [1, 2].

Theoretical research in the late 1960s and early 1970s focused on how to encode meaning and produce computationally efficient ways that the present theories of grammar could not supply. Chomsky initially presented the transformational model of language ability in 1965. Reacting to Chomsky's concepts and subsequent transformational generativist research, theories like Fillmore's case grammar, Quillian's semantic networks, and Schank's conceptual dependency theory emerged to address syntactic irregularities and semantic interpretations. Woods enhanced phrase-structure grammar through augmented transition networks, incorporating programming language elements like LISP. Distinct representational models also include Wilks' preference semantics and Kay's functional grammar [1, 2].

Several prototype models were constructed alongside theoretical research to show the usefulness of specific ideas. Any comprehensive history of NLP must men-

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tion ELIZA, an MIT Artificial Intelligence Laboratory-developed chatbot software from 1964 to 1966, conceptualized by Joseph Weizenbaum. This program, founded on the DOCTOR script, crafted by Rogerian Psychotherapists, used rule-based responses to address psychometric queries from users. Notably, ELIZA was among the select chatbots capable of passing the Turing test during that era. Another aspect of NLP that was inspired by SHRDLU, Professor Terry Winograd's Doctorate thesis, was natural language understanding. This application immersed a computer in a world of blocks, allowing it to move the blocks and respond to queries about them using natural language commands from the user. The most remarkable aspect of this system was its capacity to learn and understand with incredible precision. PARRY sought to encapsulate a paranoid hypothesis in a system. It employed groupings of keywords rather than solo keywords, and if no keywords were identified, it used synonyms. In 1978, W.A. Woods developed LUNAR, a system aimed at analyzing, comparing, and evaluating chemical data from lunar rocks and soil, a result of the Apollo moon missions. By the 1980s, computational grammar emerged as a prominent research area, focusing on meaning interpretation and user intent. The 1990s saw a rapid expansion in NLP/MT, marked by the emergence of parsers, grammars, methodologies, and practical resources related to these fields.

In the late 1970s, research shifted its focus to semantic challenges, discourse analysis, and communicative goals. Grosz's studies in task-oriented dialogues led to a thesis classifying a discourse into segments and correlating the structure of tasks with their dialogic structure. Mann and Thompson's Rhetorical Structure Theory posits a hierarchical structure in dialogues. Contributions from other researchers like Hobbs, Rosenschein, Polanyi, Scha, and Reichman were also pivotal. This era saw heightened interest in natural language generation, with McKeown's TEXT and McDonald's MUMMBLE utilizing rhetorical propositions for concise explanatory texts, often paragraph-long. It was seen as a significant success that TEXT could produce rational responses online.

The early 1980s saw a resurgence in examining nonsymbolic methods for NLP, spurred by the recognition of the inadequacy of isolated approaches and the move toward applications addressing language in a comprehensive, real-life context. The availability of essential computing resources enabled this shift. Throughout the 1980s, symbolic methods addressed major NLP concerns, while statistical approaches increasingly showed their efficacy in complementing these symbolic techniques.

NLP has progressed significantly. NLP has evolved through numerous stages, from multilingual dictionaries to handwritten rules, phrase structure to ATNs and machine learning. It is tough to deny that artificial intelligence (AI) will grow increasingly prominent in the future. There are numerous applications of NLP through the door of deep learning, and it performs as predicted by humans, if not better, in various sectors. Almost everything that humans use nowadays is automated, owing to machine learning, which has exploded in popularity with intelligent gadgets, but there is always room for development. Although it may appear like NLP has impacted every sector that might

benefit from it, there are still several areas that can benefit from this technological advancement and contribute to the development of a brighter future for everybody.

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An overview of NLP is schematically given in Figure 13.1 and a few advantages and disadvantages of using NLP are summarized in Table 13.1.

Figure 13.1: Overview of natural language processing (NLP).

Table 13.1: Advantages and disadvantages of using NLP.

Advantages	Disadvantages
The response time is fast. Usually, when a user uses a chatbot, where NLP is applied, the user gets a direct response within seconds.	Developing a new model from scratch, as opposed to using a pretrained one, might require up to a week to yield effective results, depending on the data volume.
Given a decent amount of relevant information, the accuracy of response using NLP is higher, as opposed to a human-operated chatbot.	It is not 100% reliable. If the words used in the question are ambiguous or poorly phrased, there is a possibility of errors in the predictions.
Processing language-based data is a lot easier as opposed to a human being, where a lot of tasks need to be implemented manually without fatigue and in an unbiased and consistent way which is not certain.	Because of its limited functions, the system is designed for a particular and specialized goal and cannot to adapt to different areas and challenges.

13.2 Levels of natural language processing

The seven phases of NLP.

13.2.1 Phonology

It is the systematic organization of sound. It is how speech sounds are translated inside and between words, when a computer must comprehend spoken language [3]. Undoubtedly, various languages have diverse sound systems. In phonological analysis, there are three types of rules:

13.2.1.1 Phonetic rules

It describes the sound within words.

13.2.1.2 Phonemic rules

These are used to account for differences in pronunciation when two or more words are pronounced together.

13.2.1.3 Prosodic rules

These are used to assess stress and intonation changes over a sentence. In an NLP system processing spoken input, sound waves are analyzed, transformed into digital signals, and then decoded based on a set of rules or by comparing them with the current language model.

13.2.2 Morphology

After receiving information, the initial level of analysis is morphology. It examines how words are broken down into their constituents and how this impacts their grammatical status. Morphology is mostly used to identify the elements of speech in a phrase and the words that interact with one another [3]. Morphology is the systematic description of natural language words. It defines a collection of relationships between surface forms of words and lexical forms. The surface form of a word is its spoken form, whereas the lexical form is a breakdown of the word into its lemma and grammatical descriptions. Inflectional morphology is the terminology for this job. Knowing what part of speech a word belongs to is crucial for determining its grammatical context. The data collected during the morphological step prepares the data for the syntactical stage, which examines the target language's grammatical structure very closely [4].

13.2.3 Lexical analysis

It requires identifying and analyzing the text's structure and elements of speech. The lexicon is a collection of words and phrases in a language. The process of breaking down a huge number of words into paragraphs, phrases, and words is known as lexical analysis. At this level of linguistic processing, the lexicon of a language, a collection of separate lexemes, is utilized. The lexeme plays a vital role in conveying lexical

meaning, serving as an abstract entity in morphological analysis that embodies a range of forms or "senses" linked to a solitary morpheme [3]. The key word here is "play." I enjoy playing with my little brother. The word "play" is used as a verb in this statement. We are going to the theatre tonight to see a play. In this statement, play is a noun. As a result, the word "play" may be used as a noun or a verb in a context. The determination of its part of speech and lexical meaning depends on the phrase or sentence context. This understanding forms the initial phase in advancing toward a sophisticated information retrieval system, which employs part of speech tagging to enhance precision [4].

13.2.4 Syntactic

The syntax is the process of determining the purpose for every word in a phrase and organizing the material into a more logical structure using the norms of the target language's grammar. The data may be easily altered for future investigation. Semantics is the study of words. Words and phrases are examined for their meaning [3]. At this processing stage, a representation (potentially limited) of the phrase is produced, showing the structural dependencies among words. The choice of grammar influences parser selection, making NLP not universally suitable. Consequently, persistent issues like prepositional phrase attachment parsing and conjunction scoping become less problematic when phrasal and clausal interdependence are sufficient. For most languages, syntax transmits content since order and dependence play a role in meaning [4].

13.2.5 Semantic

Semantic processing examines the interaction between word-level meanings in a phrase to identify the various meanings. Processing at this level often involves clarifying the meanings of words with multiple interpretations, similar to how words with various parts of speech are syntactically distinguished at the syntactic stage. The semantic analysis captures the meaning of a document by taking into consideration context, sentence logical structure, and grammatical roles [3]. The following are the major elements of semantic analysis:

13.2.5.1 Hyponymy

It may be described as the connection that exists between one generic word and its examples. Hypernym is the general phrase, whereas hyponyms are the specific occurrences. Color, for example, is a hypernym, whereas black, violet, and other colors are hyponyms [4].

13.2.5.2 Homonymy

It is a lexical word that refers to two or more lexical concepts that have the same spelling but have entirely different meanings. For example, the word “bat” might imply either “the bat struck the ball” or “a bat flew right across me”, with the same spelling but distinct meanings; thus, the word “bat” is a homonymy [4].

13.2.5.3 Synonymy

Synonymy occurs when two or more lexical phrases that may be spelt differently have the same or similar meaning, for instance, (ability, capacity), (instance, case), and so on [3].

13.2.5.4 Antonymy

Antonymy denotes a pair of lexical phrases with opposing meanings that are symmetrical to a semantic axis. Examples are (tall, short), (hot, cold), and so on [3, 4].

13.2.5.5 Polysemy

Polysemy is a lexical phrase that has the same spelling but many definitions that are closely related, for instance, (flow, run), (open, available), and so on [3, 4].

13.2.5.6 Meronomy

A connection in which one lexical word is a part of a bigger entity is referred to as meronomy, for instance, the word ‘toe’ is a meronym for the word ‘leg’ [4].

13.2.6 Natural language semantic analysis may be divided into two categories

13.2.6.1 Lexical semantic analysis

Lexical semantic analysis entails deciphering the content of each specific word in the text. It simply means that a word in the text has been delegated to carry the dictionary. Words, sub-words, affixes (sub-units), compound words, and idioms are all included [4]. All words, sub-words, and other lexical elements are referred to as lexical

items. In basic terms, lexical semantics is the study of the link among lexical elements, sentence meaning, and phrase structure. The following are the stages that must be followed while doing lexical semantics:

- Lexical item classification
- Lexical item decomposition
- There are also comparisons and differences between distinct lexical-semantic structures [4]

13.2.6.2 Meaning representation

Semantic analysis generates a description of a sentence's meaning. Meaning representation is the formal system that is utilized to interpret the meaning of a text. The semantic form may be used to deliberate what is true in the world as well as to extract information from it. It is critical to comprehend the building components of such forms to achieve meaning representation in semantic analysis [5]. The following building elements are crucial:

- Entities – It depicts individuals such as a specific person, place, etc. For example: Rakesh, Rajasthan, India, etc.
- Concepts – It depicts the broad category of entities such as a person, a country, and so on.
- Relations – It refers to the relationship that exists among entities and concepts. Russia, for example, is a country.
- Predicates – The verb structures are represented by predicates. Predicates include concepts such as semantic roles and case grammar.

Therefore, meaning representation demonstrates how to put the pieces of a semantic system together. To put it another way, it demonstrates how to combine entities, concepts, relations, and predicates to represent a scenario. It also makes it possible to deliberate about the semantic world [5].

13.2.7 Meaning representation methodologies

For the representation of meaning in semantic analysis, the following methodologies are used.

- First-order predicate logic (FOPL)
- Semantic nets
- Frames
- Conceptual dependency (CD)

- Rule-based architecture
- Case grammar
- Conceptual graphs

13.2.8 Discourse

NLP's discourse level works with text units bigger than a sentence, whereas syntax and semantics deal with sentence-length units. In other words, it does not perceive a series of sentences of texts as a sequence of connected sentences, each of which may be read alone. Discourse, on the other hand, concentrates on the general features of the text, which communicate meaning by connecting component sentences. This stage encompasses various discourse-processing forms, notably including the resolution of anaphoras and the identification of discourse or textual structures. At this stage, anaphora resolution is achieved by identifying the specific entity that an anaphor references. The meaning of a sentence is determined by the sentences that came before it as well as the meaning of the phrases that come after it. The basis of the discourse interactions among sentences in linked text or the discourse structure. Anaphora resolution is the process of reinstating semantically unoccupied terms, including pronouns, with the right thing to which they refer. The roles of sentences in the text are determined by discourse/text structure recognition, which contributes to the text's meaningful representation. When attempting to train chatbots to communicate successfully with people and be understood, a lot of discourse is utilized. If you tell a chatbot on a skincare website that you are searching for a nice sunscreen, the bot is not going to be very helpful if she asks what your favorite movie is. The word "that" in the phrase "He desired that," for example, is dependent on the preceding discourse context. Further approaches of NLP that involve conversation include

1. Which references are about which entities?
2. Classifying the text.
3. Grading of text quality and coherence [4].

13.2.9 Pragmatic

Pragmatic refers to anything that is practical or rational. When someone refers to you as pragmatic, they are referring to your tendency to think in terms of the practical or logical rather than the ideal scenario. Language processing at the pragmatic level involves applying real-world knowledge and grasping its impact on the transmitted content. A fuller representation is formed through the analysis of the contextual aspects in documents and queries. The aim is to elucidate how texts convey additional meaning that is not explicitly embedded within them. It thus needs considerable global

knowledge, along with a comprehension of intents, plans, and ambitions. Knowledge bases and inferencing modules may be used in certain NLP applications [4].

13.3 Applications of NLP

The major applications of NLP are detailed below:

13.3.1 Sentimental analysis

Many types of businesses utilize digital media to collect the views, wants, and intentions that people post on social media. NLP is the best method to grasp the language used and discover the meaning behind it; thus, really listening to a customer's voice demands a thorough comprehension of what they have conveyed in natural language. People frequently regard the mood (positive or negative) reflected in social media comments as the most crucial value. In reality, emotions provide a more comprehensive collection of facts that impact and, in some cases, dominate consumer decisions. As a result, NLP is extremely useful for emotion-based sentiment analysis. To improve customer experience, quickly manage customer complaints, and adjust their market position, organizations can use NLP for speech analysis, combined with a comprehensive social media monitoring strategy, to evaluate consumer sentiments and respond appropriately [6].

However, without NLP and access to the relevant data, it is impossible to find and gather the information needed to make business choices. Speech analysis is made simpler using NLP. When a consumer express concerns about a product or service via email, an NLP system can detect emotions like anger, disappointment, or frustration, categorizing the email for quick automated replies or forwarding it to the relevant team member. In a parallel scenario, financial service companies might use NLP to gauge sentiments in stock-related articles or to scrutinize reports for evaluating stock performance, aiding in decisions about buying or selling [6].

13.3.2 Text classification

Text clarification involves segmenting text into specific terms. Through text categorization, NLP can be applied to autonomously analyze text and categorize it under predefined labels or groups, based on the textual context. It is an essential component of NLP. Text is all around us in the digital world: on our social media accounts, in advertising, on webpages, in eBooks, and so on. Because the bulk of this textual data is unstructured, categorizing it may be incredibly beneficial [7]. Text classification could be divided into three main methodologies:

13.3.2.1 Rule-based system

To categorize text, these methods use handmade linguistic rules. Making a list of terms linked to a specific column and then judging the text based on the frequencies of these words is such a technique to organize content.

13.3.2.2 Machine system

To determine categories of fresh text, we may use machine learning to train models on enormous collections of text data. To train models, we must first convert text input into numerical data, a process known as feature extraction. Bag of words and n-grams are two important feature extraction approaches.

13.3.2.3 Hybrid system

These techniques are a hybrid of the two techniques discussed above. They create a classification that can be fine-tuned in specific settings using both rule-based and machine learning approaches.

Text classification may be used in a variety of ways. The following are some examples of popular applications:

- Detecting spam in emails
- Online review sentiment analysis
- Documents, such as research articles, are subjected to topic tagging
- Language recognition, similar to Google Translate
- Anonymous individuals' age and gender are determined
- Contents tagging on the internet
- Virtual assistants such as Siri and Alexa employ speech recognition

13.3.3 Chatbots and virtual assistants

For autonomous question responding, chatbots as well as virtual assistants are utilized. They are meant to comprehend natural language and engage suitably utilizing natural language generation. The use of NLP enables computers and algorithms to comprehend human interactions in a number of languages. In order to process and assess large volumes of natural language data, AI requires NLP. NLP helps AI chatbots grasp the complexity and the undertones of human conversations [8]. These chats could take place over text or the phone. While interacting with humans from all over the world, chatbots must be able to understand and emulate human discourse. Chatbots are a very recent notion and despite having a great variety of programs and NLP

tools, we basically have just two broad kinds of chatbots based on the NLP technology that they utilize [8]. These two sorts of chatbots are as follows:

13.3.3.1 Scripted chatbots

Scripted chatbots are classed as chatbots that use pre-determined scripts that are developed and saved in their library. Whenever a user inputs a question or mouths a query, the chatbot answers this inquiry as per the predetermined response, stored inside its library. One of the downsides of such a chatbot is that the user must offer their query in a structured format with comma-separated phrases or other forms of a regular expression, which allows the bot to do string analysis and interpret the query more easily. This makes this form of chatbot challenging to combine with NLP-aided speech-to-text conversion modules. Hence, these chatbots are unlikely to be developed into smart virtual assistants [9].

13.3.3.2 Artificially intelligent chatbots

AI chatbots are built to replicate human-like behaviors and answers. NLP is significantly critical to enabling such chatbots to recognize the dialects and nuances of human discourse. NLP, mixed with artificial intelligence, enables a really intelligent chatbot that can answer sophisticated queries and adapt from every conversation to deliver better-suited replies the next time. AI chatbots have been built to aid human users on diverse platforms such as autonomous chat assistance or virtual assistants, aiding in selecting music or picking a restaurant [9].

13.3.3.3 Text extraction

Automated text extraction identifies specific details within text, such as individuals, businesses, and locations, known as entity recognition. It is possible to extract key terms and preset features like product serial numbers and models from documents. Text extraction tools also facilitate the scanning of incoming support tickets to discern specific details like order numbers, brand names, and email addresses, negating the need to manually review each ticket [10]. Furthermore, text extraction can be employed for efficient data entry, allowing for the automatic extraction and system integration of the required data. As illustrated by this freeware NLP method, keyword extraction provides an outline of the structure of a document [10]. Together with keyword extraction and sentiment analysis, you may get an additional layer of information by learning which phrases consumers used most frequently to express discontent with a service or product. Text extraction, alternately referred to as keyword extraction, works by scan-

ning content, identifying useful words, and analyzing specific or essential terms/ expressions from unstructured information like news items, blog posts, surveys, and customer service requests. Text extraction is used for the following:

Extract entities
A subtask of keyword extraction is entity recognition, which is used to pull out key data elements like individuals, corporations, and email accounts to automatically update spreadsheets or databases.

Extract information
You could also give extra sets of data that have to be fetched, including product specifications (number, capacity, appearance, branding, dimension, quality, etc.).

Extract keywords
You could retrieve key phrases from the source text by using the keyword extraction model, which quickly determines that this consumer tweet is about a consumer order, and then proceed to retrieve the most essential words and phrases about the consumer experience.

Summarize a paragraph or text
Identifying the most significant terms/expressions from a dataset, social media posts, emails, and other sources is another method to utilize text extraction [10]. It can accurately condense thousands of customer reviews and large files by identifying the most utilized and most significant terms and expressions.

Clean data
These are software like email cleaners, which remove autographs, nondisclosure agreements, contact terms, legal agreements, and prior statements within an email thread, so that you eventually wind up with a “clean” variant using simply the newest response. Text extraction is also beneficial for “cleaning” words and sentences to eliminate superfluous data that can modify the precision of one’s findings [10].

Machine translation
Machine translation or MT is a method where a computer software translates source text from one language to another without human involvement. At its base level, machine translation conducts a direct replacement of words from one language with those from another using just the atomic terms [11]. There are four types of machine translation:

Statistical machine translation (SMT)

It makes use of statistical models that are based on the evaluation of enormous volumes of linguistic data. It also intends to determine a comparison of a term from the original language and a term from the objective language. An actual example of this would be Google Translate [11]. Currently, SMT is outstanding for basic translations; however, its most significant drawback is that it does not take in the context, which means translation can routinely be inaccurate or the quality of translation is not great.

Rule-based machine translation (RBMT)

The basics of grammatical structure are effectively translated by RBMT. It conducts a grammatical evaluation of the original language and the desired language to generate the translated text. But, RBMT involves comprehensive editing, and its high dependence on dictionaries means that expertise is acquired after a significant duration.

Hybrid machine translation (HBT)

HMT, as the word shows, is a hybrid of RBMT and SMT. It employs a translation memory, making it certainly more effective regarding quality [11]. Yet, even HMT has several drawbacks, the most significant of which is the requirement for large editing, so human translators will also be necessary.

Text summarization

This process involves automatically condensing text by selecting the most crucial information. Its significant intention is to facilitate the method involved with glancing through huge volumes of information, like logical distributions, news data, or lawful desk work. The approach in which a software program reduces lengthy texts and provides summary to convey the intended meanings is known as automatic text summarization and is a frequent challenge to machine learning and NLP [12]. It is the method involved with developing a more limited, rational, and smooth synopsis of an extended text report and incorporates the featuring of the text's significant points. Text identification, interpretation and summary production, and examination of the resulting summary are some of the primary issues addressed in the process of text summarizing [13]. The important tasks in extraction-based summarizing include recognizing key terms in the text and applying them to identify significant information to be included in the summary. NLP can be used for summarization in two primary forms: extraction-based, which involves pulling out key phrases to provide a summary without extra information, and abstraction-based, which generates new phrases that rephrase the original content. The latter approach is more commonly used and tends to be more effective [12, 13].

Extractive summarization

Extractive approaches aim to summarize articles by finding the relevant statements or terms from the actual text and stitch together parts of the information to generate a shortened version. These picked phrases are then utilized to build the summary.

Abstractive summarization

This approach, unlike extraction, depends on being able to paraphrase and reduce sections of a paragraph through powerful natural language capabilities since abstractive machine learning systems can develop new words and sentences to capture the meaning of the original content [13]. When such abstraction is performed properly in deep learning issues, they may aid in overcoming grammatical inaccuracies.

Market intelligence

NLP assists advertisers with looking further into their clients and utilize that data to plan a more productive approach. Examining unstructured information for subjects, mindset, catchphrases, and reason might assist you with fundamentally further developing statistical surveying by revealing insights into patterns and business, opening doors [14]. One could likewise utilize information investigation to distinguish client issues and consistently watch out for opposition. In the analytics industry, there is a large selection of tools and KPIs to pick from in order to measure the marketing data. One rising use case is Normal Language Processing (NLP), the method involved with getting a PC to understand and analyze data involving human language. NLP is commonly used to get qualitative knowledge of the "why" and "what" of a situation and helps users to make more informed judgments. In marketing analytics, NLP may be used to understand the audience's intentions so that one can design better, more efficient marketing tactics [14]. For the NLP's capacity to evaluate – and even forecast – language inputs provide a wide variety of information about users' traits, interests, requirements and preferences. To digital marketers, this is essential information that recognizes a corporation's target audience and helps improve marketing techniques to turn them into paying clients. In this approach, NLP helps digital marketers accomplish various objectives, whether generating more valuable content or identifying the finest keywords [14].

Market intelligence autocorrect

Grammar checking software and autocorrect capabilities rely heavily on NLP. For instance, Grammarly employs NLP to enhance your writing skills by detecting errors in grammar, spelling, and sentence construction [15].

Market intelligence intent classification

The expectation or reason that supports a text is recognized through intent classification. Beyond chatbots, intent classification is valuable in sales and customer service areas. By examining customer communications, including emails, chats, and social media engagements, it is possible to pinpoint potential buyers. Prompt identification and qualification of these prospects increase the likelihood of converting them into clients. The email classifier categorizes responses into segments such as Likely, Unlikely, and Unsubscribe [16]. Ultimately, checking for client intention in support requests or any digital platforms could caution you about clients who will stir, permitting you to mediate with a reengagement strategy.

Market intelligence urgency detection

Text can also benefit from NLP approaches for detecting urgency. You can customize an urgency detection algorithm to recognize specific words and phrases signaling urgency or dissatisfaction. This approach helps prioritize essential requests, preventing them from getting lost among numerous unresolved tickets [17]. Urgency detection aids in the improvement of competence, reaction times, and efficiency, resulting in customer satisfaction.

Market intelligence speech recognition

NLP utilizes speech recognition technology to transform spoken language into a format readable by machines. Speech recognition is integral to virtual assistants like Siri, Alexa, and Google Assistant [18]. The commercial applications of voice recognition are expanding, such as integrating speech-to-text features in business software for automated call transcriptions, email sending, and document translations [18].

13.4 Applications of NLP in healthcare sector

The major applications of NLP in healthcare sector are detailed in this section.

13.4.1 Documentation

The clinical documentation produced by NLP frees physicians from the time-consuming physical systems of EHRs, allowing them to dedicate more time to the patient. Documents can be created on the go using a combination of speech-to-text and formulated data entry [19].

13.4.2 Computer-assisted coding

CAC collects information about procedures and therapies to grasp every potential code and maximize claims. Although it is one of the most popular applications of NLP, only 30% of people utilize it. It has improved coding speed but has fallen short on accuracy [20].

13.4.3 Clinical trial matching

A prominent use case is in healthcare, where NLP and computers are used to recognize patients for clinical trials. Several firms are using NLP in healthcare engines for trial matching to address the issues in this field. With recent advancements, NLP can automate trial matching and make it a painless process. IBM Watson Health and Inspirata are two examples of clinical trial matching use cases, both of which have invested significant resources in using NLP to assist cancer studies [21].

13.4.4 Prior authorization

According to research, payer prior authorization rules for medical workers are only getting more stringent. These demands raise practice expenses and cause delays in patient service. Because of NLP, the issue of whether payers will agree and enact compensation may not exist in the future. IBM Watson and Anthem have already developed an NLP module utilized by the payer's network to quickly determine previous authorization [22].

13.4.5 Chatbots

While there is now no such solution, speech recognition software has a strong chance of supporting individuals in updating clinical documentation. This is where Amazon's Alexa or Google's Assistant comes in handy. Microsoft and Google have partnered up to accomplish this task. As a result, Amazon and IBM are expected to follow suit. Chatbots or virtual personal assistants are employed in a number of ways in today's digital world, and the healthcare industry is no exception. These assistants may now capture symptoms and refer patients to the most appropriate practitioner. Smart Exam is a "virtual medical assistant" developed by BRIGHT.MD, a new chatbot company that leverages conversational NLP to collect and compare personal health data [23].

13.4.6 Risk managing

Condition of hierarchy: The original goal of category coding, a risk adjustment methodology, was to estimate future patient care expenses. HCC coding will become more prominent in value-based payment structures. ICD-10 coding is used to assign risk ratings to each patient in HCC. NLP can assist in assigning a risk factor to patients and use their scores to estimate healthcare expenses [24].

13.4.7 Review management

NLP may also help manage internet reviews in the healthcare business. It can gather and evaluate hundreds of healthcare reviews from third-party listings daily. PHI (protected health information), vulgarity, and other data related to HIPAA compliance can also be detected using NLP. It can also calculate human emotions and the context in which they are manifested, in a short time. Some systems may even listen to the customer's voice in reviews. This can help the physician better understand how patients talk about their care and communicate more effectively using standard terms and phrases. NLP can track consumer attitudes by interpreting positive and negative terms in reviews [25].

13.4.8 Root cause analysis

NLP can also aid in managing online reviews in the healthcare industry. Every day, it can collect and analyze hundreds of evaluations on healthcare from third-party listings. NLP also detects PHI, profanity, and other data relevant to HIPAA compliance. It may also quickly assess human emotions and the environment in which they are expressed [26].

13.5 NLP-based approaches in healthcare sector-review

An overview of NLP in healthcare is shown in Figure 13.2. A few well-known NLP-based approaches in healthcare sector, along with the methodology, advantages and disadvantages of those methods, are summarized in Table 13.2.

A few other relevant works in the domain of NLP for healthcare sector are discussed below.

The authors developed an artificial intelligence SERA algorithm that predicts and diagnoses sepsis using both structured and unstructured clinical information [32]. The

Figure 13.2: NLP in healthcare.

Table 13.2: Advantages and disadvantages of using natural language processing (NLP).

Scheme	Methodology/approach	Advantages	Disadvantages
Validation of prediction models for critical care outcomes using natural language processing of electronic health record data [27]	This chapter analyzed the power shift for a mortality prediction model among ICU patients, gained by combining clinical trajectory metrics along with NLP of clinical text, and assessed the generalizability of this methodology	Can be adapted to EHRs used by healthcare researchers and practitioners for a number of objectives, such as risk adjustment in clinical trials and quality improvement	Were not able to directly compare their models; this restricted the models to the most prevalent terms; their applicability to any particular hospital is not assured
How to automatically turn a patient's experience from free-text responses into actionable insights: a NLP approach [28]	In order to accelerate performance, frequency and emotion metrics are measured	The model utilized is portable even if it collects a similar collection of data topics from multiple hospitals	Precise opinion estimates from patient responses can be obtained by using multi-labeling to capture several subjects
Identification of suicidal behavior among psychiatrically hospitalized adolescents using language processing and machine learning of electronic health natural records [29]	This paper proposes a classification model using the random forest ML technique, in which the terms that were related to the outcome of the attempt at suicide were discovered using NLP and were enhanced to denote recognized risk and protective variables for a teen suicide attempt	It is the first instance of unstructured datasets being examined employing NLP to create a classification model in a patient sample of adolescents	Restricted by a limited dataset, since the study does not provide information on the individual who attempted suicide, the dates, etc. and one cannot tell if the algorithm is categorizing already occurring incidents or forecasting upcoming ones

Table 13.2 (continued)

Scheme	Methodology/approach	Advantages	Disadvantages
Attention neural network for biomedical word sense disambiguation [30]	The research suggests a disambiguation technique using the attention implementation in neural networks This utilizes biomedical text	It simultaneously includes data about words' prefixes and suffixes This is used to determine the uncertain term in its context	When biomedical data loss happens, the accuracy of the outcomes can be increased by using unsupervised learning algorithms and unlabeled datasets
A NLP-based approach for identifying hospitalizations for worsening heart failure within an integrated healthcare delivery system [31]	This research examines the total burden of worsening heart failure (WHF) hospitalizations as well as the temporal patterns in the rate of hospitalization	The number of WHF hospitalizations recognized by NLP-based algorithms grew from 5.2 to 7.6.	Several possible treatment strategies for WHF are not technically viable This research additionally removed data further than the first 72 h of admission to distinguish between hospitalizations for WHF and protracted hospitalizations exacerbated by WHF

SERA algorithm is meant to operate in the background in the initial state. The program can be set to run instantaneously after a doctor changes clinical information in the electronic medical record system in the second mode, in an ad hoc manner. With independent clinical notes, this algorithm exhibited good prediction accuracy 12 h before sepsis started, with AUC 0.94, sensitivity 0.87, and specificity 0.87. They analyzed and compared the SERA algorithm to physician predictions and discovered that it has the potential to improve early diagnosis of sepsis by up to 32% and minimize false positive results by up to 17%.

NLP to extract information from multiple unorganized health records created in the predictive stage identification of women with breast cancer [33] – The authors developed a principle system and an ML technique for more accurately predicting the prognostic stage. The study relied on data from clinical, pathology, and immunohistochemical evaluations from two medical institutions in urban and remote areas. The average prognosis stage accuracy in rural and urban areas is 92% and 82%, respectively. This study attained an average prognosis stage accuracy of 87% in both locations.

In another research, the authors manually annotated a collection of mental health EHRs with illness-beginning mentions, modeled the foundational NLP challenge as a paragraph classification technique in which files were segmented into paragraphs utilizing HTML tags, and integrated numerous occurrence paragraphs just at

the patient level to provide a sorted list of possible diseases onset dates [34]. The proper onset date was discovered in the top three NLP predictions in 22 of 31 test subjects (71%). The suggested method was also used on a large scale, allowing for the estimation of an onset date for 2,483 cases.

The significance of EHRs that use NLP and deep learning is discussed in [35]. In the health sector, NLP would play an essential role in speeding decision-making. The true benefits of establishing effective algorithms, however, will be strongly reliant upon the quality of the data you capture and preserve. Specialists will be able to concentrate on providing patients with more value-added treatment as a result of the speedier decision-making procedure.

A comprehensive overview of the difficult difficulties involved in designing suitable intrinsic and extrinsic assessment methodologies for NLP research that will be utilized for health outcomes studies and vice versa [36]. A particular emphasis is made on mental illness study, which is still underexplored by the medical NLP industry but in which NLP methodologies are beneficial. They have identified three primary obstacles: data accessibility, assessment workbenches, and disclosure requirements from a healthcare and NLP standpoint. Based on those, they also provided concrete advice for every highlighted challenge.

13.6 Research challenges in the NLP domain

Despite recent advancements, there are still obstacles to the mainstream adoption of NLP technologies.

Like other AI approaches, NLP is highly reliant on the training data's accessibility, reliability, and kind. For training and implementing robust NLP models, having access to and availability of suitably annotated datasets is critical. For example, unfettered accessibility to a publisher or primary hospital data is required to develop and use algorithms capable of comprehensively synthesizing published studies on a specific topic or a study and data extraction from EHR systems [37]. While the amount of openly available biological databases and pretrained algorithms has grown throughout the years, one is concerned with public health-related issues that are still scarce.

Another critical aspect of training and deploying NLP techniques in health care contexts is the ability to de-bias data. Failure to account for biases in the NLP building of the model, implementation, and evaluation could degrade the model's outputs and worsen existing health disparity. It is crucial to highlight, however, that although datasets and evaluations are modified for biases, an equal influence across ethically relevant strata cannot be guaranteed. For example, when using health data available across social media sites, certain age and socioeconomic categories must be considered. A monitoring program trained on Facebook data is more likely to favor healthcare data and language peculiarities unique to an older demographic than one trained using

Snapchat data [37]. In response to government and academic attempts to prevent unacceptably advanced AI, various model agnostic techniques have recently been introduced to examine and remedy unfairness in ML models.

Limited data access is now one of the most significant roadblocks to the continued expansion of NLP systems in public health. Health data is mainly managed regionally, and there is a hesitation to offer unrestricted system access and its linkage with other databases because of security and confidentiality issues. There are also difficulties with the general sentiment of data privacy and security. A recent poll of people on social media found that the most percent viewed monitoring of personal social media data to discover mental health problems as "invasive and exposing" and that they would hardly consent to it [37].

Before essential NLP public health tasks, such as real-time monitoring of national disease trends, can be implemented at scale, governments will need to agree on a realistic extent and accessibility of health-relevant data sources. Future uses of NLP in the study of personal health information will be dependent on the capacity to incorporate differential privacy within models, during both training and after implementation, in order to avoid privacy breaches and data exploitation [37]. The present means for obtaining full-text publications also limit access to vital data. Full automation of PICO-specific knowledge extraction and synthesis will necessitate unlimited access to publications or new data storage formats.

Finally, like with any technological advances, NLP models must be assessed and evaluated to ensure that they perform as planned and keep up with evolving ethical attitudes in society. These NLP technologies must be evaluated to ensure that they work as predicted and that bias is considered. Although many techniques now achieve human-like or better results on textual analysis tests, it is critical not to confuse high scores with actual language comprehension. It is also vital not to mistake a lack of real language comprehension for a lack of utility [37]. With the rising availability of labeled data, models with a "very weak" depth of comprehension can still be highly effective at extracting information, classification, and prediction tasks.

13.7 Conclusion

NLP is a method of analyzing texts using technology. The NLP entails acquiring information on how humans comprehend and utilize language. This is done to create suitable tools and procedures that will allow computer systems to interpret and modify natural languages to complete various jobs. Health NLP is getting prominence, given its critical function in a wide range of methodological development and applications. The amount of data produced by the healthcare industry is far too large presently, and manually analyzing it all is a time-consuming operation. Linguistics and computer science are both subfields of NLP that will aid in the processing and analysis of

unstructured natural language claims for a variety of purposes. This chapter gives an overview of the evolution of NLP and discusses the levels of NLP in detail. Furthermore, the applications of NLP in healthcare, as well as other sectors, are highlighted. This paper discusses the numerous well-known NLP-based methodologies available for processing healthcare information and a brief outline of the challenges in this domain that require future attention. This paper aims to inspire scholars interested in utilizing NLP in various sectors, especially in healthcare.

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Authors' biography

Shashank Kamineni is a dynamic and driven computer science professional, whose journey in the field began at SRM University AP, where he completed his B.Tech. His early fascination with artificial intelligence and machine learning was sparked by a childhood interest in video games, which later evolved into a deeper engagement with these cutting-edge technologies. Currently, Shashank is furthering his expertise by pursuing a master's degree in computer science at George Mason University in the USA. His academic and practical pursuits in AI and ML have positioned him at the forefront of innovation in these areas, reflecting his commitment to being at the cutting edge of technological advancements.

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Tejodbhav Koduru is an emerging researcher in the field of computer science, currently advancing his studies with M.Sc. in computer science at the University of Illinois at Chicago. His research interests are rooted in data analytics, machine learning, and wireless sensor networks, with a specialization in big data analytics. Tejodbhav has gained practical experience as a research intern at both Trellis Labs P Ltd and SRM University – AP, where he worked on projects involving the detection and classification of Asperger's syndrome and predicting the prices of virtual machine instances using ARIMA and LSTM models. He has contributed to several peer-reviewed publications, demonstrating his commitment to advancing knowledge in wireless sensor networks and IoT. His notable achievements include receiving two gold medals at Research Day 2022 SRMAP for his contributions in the domains of IoT and AI/ML. Tejodbhav's technical acumen, combined with his dedication to research, positions him as an insightful contributor to the evolving landscape of computer science and technology.